Lecture 21

Determinants and invertibility

Uses.

Summary 2: Finite sums can be regrouped, and induction proves statements for every positive integer size.
Summary 17: Elementary row operations reduce any matrix to echelon and reduced echelon form. Scaling operations divide only by nonzero scalars.
Summary 19: An operation is left multiplication by its elementary matrix. Products represent composition and are associative.
Summary 20: The determinant is defined by first-column expansion. Upper triangular determinants are diagonal products and $\det I_n = 1$. A square matrix is invertible exactly when its reduced form is I_n.

Objectives. *The determinant's recursive definition gives rules for elementary row operations. These rules turn elimination into a determinant calculation and prove that the determinant is nonzero exactly when a square matrix is invertible. Applying the same operations to a product proves multiplicativity.*

By the end of this lecture you will be able to:

- Prove the determinant's row linearity and row-operation rules from first-column expansion.
- Compute a determinant by elimination while recording every scale and sign.
- Prove the equivalence between a nonzero determinant and invertibility.
- Prove multiplicativity and derive its inverse and real-volume consequences.

21.1 Linearity in one row

Keep $K = \mathbb{R}$ or $K = \mathbb{C}$ and all matrices square of size $n \geq 1$. We use the determinant defined in Lecture 20, not a new definition by row rules. For rows written horizontally, let $D_r(u)$ denote the determinant with row r equal to $u \in K^{1\times n}$ and every other row fixed. The determinant is *linear in row r* if

$$D_r(cu + dv) = cD_r(u) + dD_r(v) \qquad (u, v \in K^{1\times n},\ c, d \in K). \tag{21.1}$$

Linearity holds separately for each row. Other rows are held fixed when this identity is used. It does not assert that the determinant is linear as a function of the whole matrix.

For a concrete first-column expansion, vary the second row (u_1, u_2, u_3):

$$\det\begin{pmatrix} a & b & c \\ u_1 & u_2 & u_3 \\ d & e & f \end{pmatrix} = a\det\begin{pmatrix} u_2 & u_3 \\ e & f \end{pmatrix} - u_1\det\begin{pmatrix} b & c \\ e & f \end{pmatrix} + d\det\begin{pmatrix} b & c \\ u_2 & u_3 \end{pmatrix}.$$

The middle minor deletes the varying row, leaving a fixed determinant multiplied by the varying entry u_1. The other minors retain the varying row, with its first entry removed. This is the distinction used in the induction.

Theorem 21.1 (Row linearity). *The recursively defined determinant is linear in each row.*

Proof. We argue by induction on n. At size 1 the determinant is its sole entry, which is linear in that entry. Assume row linearity has been proved for size $n-1$ and fix a row index $r \in \{1, \dots, n\}$ at size $n \geq 2$. Expand the determinant along column 1, as in Eq. (20.2).

In the term with index $i = r$, the minor deletes the varying row. Its determinant is therefore fixed, and the coefficient in column 1 is $cu_1 + dv_1$. This term splits into c times its value for row u and d times its value for row v.

In each term with index $i \neq r$, the first-column coefficient belongs to another row and is fixed. The minor contains the varying row with its first entry removed. That minor row is $c(u_2, \dots, u_n) + d(v_2, \dots, v_n)$. By the induction hypothesis, the minor determinant splits into the same linear combination of the two corresponding minor determinants. The fixed sign and first-column coefficient multiply both parts.

Every term of the finite expansion therefore has the required split. Adding them gives Eq. (21.1) at size n. This completes the induction for every row and every size. ∎

A zero row makes a determinant zero: in the linearity formula choose $c = d = 0$. Scaling a single row by $c \in K$ scales the determinant by c, by choosing the other scalar to be zero. Scaling every row once gives, for $A \in K^{n \times n}$,

$$\det(cA) = c^n \det A.$$

This includes $c = 0$ because $n \geq 1$. The exponent counts the rows scaled; it is not generally just one factor of c.

21.2 Equal rows, swaps and additions

First we prove the vanishing result needed for row swaps.

Lemma 21.2 (Equal adjacent rows). *If two adjacent rows of a square matrix are equal, its determinant is zero.*

Proof. At size 2 the formula is $ad - cb$; for equal rows (a, b) and (a, b) it is $ab - ab = 0$. Assume the result holds at size $n-1$ and consider size $n \geq 3$ with equal adjacent rows r and $r+1$. Expand along column 1. If the deleted row i is neither of those rows, the minor retains them as equal adjacent rows. Its determinant is zero by induction.

There remain just the terms deleting row r or row $r+1$. Their first-column entries are equal. Their minors are identical: either deletion leaves one copy of that same row in the same position among the other rows. Their signs are $(-1)^{r+1}$ and $(-1)^{r+2}$, which are opposite. Thus these two terms cancel. The full expansion is zero. The induction proves the result; at size 1 there are no two adjacent rows. ∎

Hold every other row fixed, and put rows $u + v$ in adjacent positions. By Lemma 21.2 that determinant is zero. Applying row linearity in each of those positions expands it into four terms:

$$0 = D(u, u) + D(u, v) + D(v, u) + D(v, v) = D(u, v) + D(v, u).$$

Here $D(u, v)$ denotes the determinant with those two rows in the indicated order; the equal-row terms vanish by the lemma. Therefore exchanging adjacent rows changes the sign.

For arbitrary row positions $r < s$, exchange their rows by adjacent swaps. Move the original row r down to position s, using $s - r$ swaps. The original row s is then in position $s - 1$; move it up to position r, using $s - r - 1$ more swaps. All intervening rows return to their original positions. The total $2(s - r) - 1$ is odd, so exchanging any two different rows changes the determinant's sign. If equal rows are not adjacent, adjacent swaps can bring them together. The resulting determinant is zero by the lemma, and reversing those sign changes shows that the original determinant was zero as well.

Now replace row i by row i plus c times another row j, with $i \neq j$. Row linearity gives the original determinant plus c times a determinant in which rows i and j are equal. The second determinant is zero. This row addition therefore preserves the determinant for every $c \in K$.

Theorem 21.3 (Determinants under row operations)**.** *For a square matrix over K, exchanging two different rows negates the determinant. Multiplying one row by $c \in K$ multiplies it by c. Adding c times a different row to a row leaves it unchanged. A zero row or two equal rows makes it zero.*

All claims have been derived from the first-column definition. For reversible elimination, a scaling factor is required to be nonzero; the determinant scaling identity itself also holds when that factor is zero.

For an example that records all three changes, start with

$$M = \begin{pmatrix} 2 & 1 & 0 \\ 1 & -1 & 2 \\ 3 & 0 & 1 \end{pmatrix}.$$

Its defining first-column expansion gives $\det M = 2(-1) - 1(1) + 3(2) = 3$. Multiply row 1 by -3, exchange rows 2 and 3, and then add twice the new row 1 to row 3. The resulting matrix is

$$W = \begin{pmatrix} -6 & -3 & 0 \\ 3 & 0 & 1 \\ -11 & -7 & 2 \end{pmatrix}.$$

The scaling multiplies the determinant by -3, the swap multiplies it by -1, and the addition multiplies it by 1. Hence

$$\det W = (-1)(-3) \det M = 9.$$

The negative scaling and the row swap produce separate signs. Neither should be discarded when recording the operations.

21.3 Elimination computes the determinant

Suppose a sequence of reversible row operations takes $A \in K^{n \times n}$ to an upper triangular matrix U. Let s count the swaps and let $c_1, \ldots, c_k \neq 0$ be all row-scaling factors actually used. Row additions need no factor. Successive applications of Theorem 21.3 give

$$\det U = (-1)^s (c_1 \cdots c_k) \det A.$$

Use an empty product of 1 when there are no scalings. The diagonal-product formula of Lecture 20 therefore gives

$$\det A = \frac{(-1)^s}{c_1 \cdots c_k} u_{11}u_{22}\cdots u_{nn}. \tag{21.2}$$

The denominators are nonzero because the row operations are reversible. When elimination uses only row additions and swaps, the denominator is 1. Every square echelon matrix is upper triangular: its successive pivot columns have indices at least $1, 2, \ldots$ in their respective rows, and its zero rows lie below them. Thus Gaussian elimination always supplies such a calculation, including matrices with missing pivots.

Consider the real parameter $\lambda \in \mathbb{R}$ in

$$A_\lambda = \begin{pmatrix} 0 & 2 & 1 & 0 \\ 1 & -1 & 0 & 2 \\ 2 & 0 & 3 & 1 \\ 1 & 1 & 1 & \lambda \end{pmatrix}.$$

Exchange the first two rows. Subtract twice the new row 1 from row 3, and subtract row 1 from row 4. The matrix becomes

$$\begin{pmatrix} 1 & -1 & 0 & 2 \\ 0 & 2 & 1 & 0 \\ 0 & 2 & 3 & -3 \\ 0 & 2 & 1 & \lambda - 2 \end{pmatrix}.$$

Subtract row 2 from each of rows 3 and 4. The resulting upper triangular matrix is

$$U_\lambda = \begin{pmatrix} 1 & -1 & 0 & 2 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 2 & -3 \\ 0 & 0 & 0 & \lambda - 2 \end{pmatrix}.$$

Only one swap and determinant-preserving additions were used. Therefore the determinant is

$$\det A_\lambda = -\det U_\lambda = -4(\lambda - 2).$$

This equality holds also at $\lambda = 2$; no division by $\lambda - 2$ was used. The triangular matrix has four pivots when $\lambda \neq 2$ and three when $\lambda = 2$. In the latter case, assigning the last coordinate 4 and working backwards gives the homogeneous column $(-11, -3, 6, 4)$. Its original left sides are $-6 + 6 = 0$, $-11 + 3 + 8 = 0$, $-22 + 18 + 4 = 0$ and $-11 - 3 + 6 + 8 = 0$. The nonzero column confirms failure of invertibility at that parameter.

21.4 The nonzero-determinant criterion

Theorem 21.4 (Determinant and invertibility). *For $A \in K^{n\times n}$, the matrix A is invertible if and only if $\det A \neq 0$.*

Proof. Reduce A to reduced row-echelon form R. The total determinant factor from the reversible operations is a nonzero scalar κ, so $\det R = \kappa \det A$ by the row-operation theorem.

If A is invertible, Lecture 20's elimination criterion gives $R = I_n$. Thus $1 = \kappa \det A$, and consequently $\det A \neq 0$. If A is not invertible, its reduced square matrix has fewer than

n pivots. Since each nonzero row has one pivot, R has a zero row. Hence $0 = \det R = \kappa \det A$. The factor κ is nonzero, so $\det A = 0$. The two cases prove both directions of the equivalence. ∎

A square matrix with determinant zero is called *singular*. By the theorem this is exactly a noninvertible square matrix. A square matrix with nonzero determinant is called *nonsingular*. For the running family, the theorem gives invertibility exactly when $\lambda \neq 2$, agreeing with the pivot calculation. A zero determinant does not mean the matrix itself is zero, as A_2 shows. It records failure of invertibility, not the vanishing of every entry.

21.5 Multiplicativity from row operations

Theorem 21.5 (Multiplicativity). *For $A, B \in K^{n\times n}$,*

$$\det(AB) = \det A \det B. \tag{21.3}$$

Proof. Choose reversible row operations reducing A to R and let E be their product of elementary matrices, so $R = EA$. Let $\kappa \neq 0$ be the product of their determinant factors. The factors depend only on those operations, not on the starting matrix. Applying the same sequence to any square M gives $\det(EM) = \kappa \det M$. This follows by applying the proved row-operation rules at every step.

If A is invertible, its reduced form is I_n by Lecture 20. Apply the factor identity first to $M = A$, and then to $M = AB$. Associativity gives $E(AB) = (EA)B = B$, so

$$1 = \kappa \det A, \qquad \det B = \kappa \det(AB).$$

Solving the second equality and using the first gives $\det(AB) = \det A \det B$ in this case.

If A is singular, R has a zero row, as proved in the preceding criterion. That row of RB is zero as well, because each of its entries is a sum of zero scalar products. Thus $\det(RB) = 0$. But $RB = E(AB)$ by associativity, so $0 = \kappa \det(AB)$ and $\det(AB) = 0$. Also $\det A = 0$ by singularity, making the required right side zero. This proves the formula in the remaining case, without any condition on B. ∎

Example 21.6 (A product over $\mathbb{C}$). For the complex matrices

$$P = \begin{pmatrix} 1 & \mathrm{i} \\ \mathrm{i} & 1 \end{pmatrix}, \qquad Q = \begin{pmatrix} 1 & 1 \\ 0 & -\mathrm{i} \end{pmatrix},$$

compute the determinant of their product. Their determinants are $1 - \mathrm{i}^2 = 2$ and $-\mathrm{i}$, respectively. Multiplicativity therefore gives $\det(PQ) = -2\mathrm{i}$. The explicit product is

$$PQ = \begin{pmatrix} 1 & 2 \\ \mathrm{i} & 0 \end{pmatrix},$$

whose size-2 determinant is also $1 \cdot 0 - 2\mathrm{i} = -2\mathrm{i}$. The nonzero value proves that the product is invertible over $\mathbb{C}$. This scalar calculation does not assert a real signed-volume interpretation for the complex matrices.

For an invertible matrix A, applying the product rule to $AA^{-1} = I_n$ gives

$$\det(A^{-1}) = \frac{1}{\det A}.$$

Division is valid by Theorem 21.4. For every integer $r \geq 0$, repeated multiplication gives $\det(A^r) = (\det A)^r$, starting from $\det I_n = 1$ at $r = 0$. At that order the scalar power also means 1, including for singular A. No formula $\det(A + B) = \det A + \det B$ follows from row linearity: for $n = 2$, the matrices $A = B = I_2$ give $\det(A + B) = 4$, whereas $\det A + \det B = 2$. This guards the distinction between linearity in one row and a claim about the full matrix.

For real 3×3 matrices, multiplicativity also determines volume change. Let the columns u, v, w describe an ordered parallelepiped. The columns after the linear map $x \mapsto Ax$ are Au, Av, Aw. The definition of matrix multiplication gives $[\, Au \;\; Av \;\; Aw \,] = A[\, u \;\; v \;\; w \,]$. Consequently their signed volume is

$$\det[\, Au \;\; Av \;\; Aw \,] = \det A \det[\, u \;\; v \;\; w \,].$$

Taking absolute values gives the ordinary-volume factor $|\det A|$, including zero for a singular map. For a nonzero original volume and nonsingular A, the orientation reverses exactly when $\det A < 0$. The same product argument in two real dimensions gives the signed-area factor $\det A$ and ordinary-area factor $|\det A|$. The calculation on Sheet 20 is one such real affine example; translation cancels when forming side columns.

Summary 21.

- Theorem 21.1: the determinant is linear separately in each row, with the other rows fixed. In particular $\det(cA) = c^n \det A$ for $n \geq 1$.
- Theorem 21.3: swaps negate, row scalings multiply by the scalar, and adding a multiple of another row preserves the determinant. A zero row or equal rows makes it zero.
- Equation (21.2): elimination computes the determinant from triangular pivots with every swap and nonzero row-scaling factor recorded.
- Theorem 21.4: a square matrix is invertible exactly when its determinant is nonzero. A singular matrix has determinant zero.
- Theorem 21.5: $\det(AB) = \det A \det B$ for square same-size factors. Thus $\det(A^{-1}) = 1/\det A$ when A is invertible. For real maps, ordinary area or volume scales by $|\det A|$.

Sheet 21 applies the row rules and product law to new data and exceptional cases. Lecture 22 develops expansion along other rows and columns and the adjugate formula.

References

D. C. Lay, S. R. Lay and J. J. McDonald, *Linear Algebra and Its Applications*, 5th ed., Pearson, 2016, §§3.1–3.3. S. Lang, *Introduction to Linear Algebra*, 2nd ed., Springer, 1986, Chapter VII.